

Scalability & Future Plan

Date	03 July 2026
Team ID	
Project Name	Predicting Human Development Index (HDI) Using Machine Learning
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the HDI Prediction System can be scaled to support a larger number of users, handle real-time global economic data, and incorporate advanced policy simulation features. It also presents the future roadmap for enhancing the system beyond its current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	The application currently runs on a local Flask development server and is not deployed to a cloud platform.	Limits accessibility to users on the local system only.	High
2	HDI prediction is based on historical socio-economic data and manually entered parameters.	Predictions do not incorporate live global indicators or real-time socio-economic changes.	High
3	The system does not support real-time data API integration, historical prediction databases, or user collaboration tools.	Policymakers cannot retrieve live country reports or save historical simulation runs.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
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1	User Load	Supports a single or limited number of users on a local Flask server.	Deploy the application on IBM Cloud, Render, Railway, or AWS to support multiple concurrent users.
2	Data Storage	Uses a CSV dataset and serialized model files (.pkl).	Integrate a relational database such as PostgreSQL or MySQL to store prediction history, user accounts, and updated country profiles.
3	Performance	Predictions are generated using a locally stored Linear Regression model.	Optimize the model pipeline, implement caching, and use production-grade servers such as Gunicorn or uWSGI for improved scalability and faster response times.
4	Security	Basic client-side and server-side input validation is implemented.	Add user authentication, HTTPS, secure REST APIs, and role-based access control to enhance security and protect user simulation data.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on IBM Cloud/Render and improve the user interface.	1–2 Months	Enable public access, improve usability, and support remote socio-economic assessment.
Phase 2	Integrate real-time world data APIs (e.g. World Bank API or UNDP Data API) to automatically populate country indicators.	3–5 Months	Increase prediction reliability and ease of use by incorporating current global metrics.
Phase 3	Implement comparative country analysis, visual trend mapping, and	4–6 Months	Provide visual policy monitoring and professional report outputs.

	automated PDF report generation.		
Phase 4	Develop mobile applications with multilingual support and expand the model to support deep learning-based forecasting.	6–12 Months	Improve accessibility and support long-term economic forecasting models.

Architecture Scaling

Cloud Deployment

- Deploy the Flask server to cloud platforms such as AWS Elastic Beanstalk or Heroku.
- Use Docker containers to ensure consistency across development and production environments.

REST API Development

- Expose the prediction module as a REST API endpoint so other tools and research portals can query predictions programmatically:

```
```json
POST /api/predict
{
 "country": "Norway",
 "life_expectancy": 83.2,
 "expected_years_schooling": 18.0,
 "mean_years_schooling": 13.0,
 "gni_per_capita": 66000
}
...`
```

### Conclusion:

The HDI Prediction System provides an accurate Machine Learning-based solution for predicting Human Development Index scores using historical indicators. Future enhancements will focus on cloud deployment, live economic API integration, relational database storage, comparative analytics, and mobile application development. These improvements will make the application more scalable, reliable, and suitable for real-world policy analysis and decision support.